

Quiz (Habanero)

Q1. Solve $|x - 2| < 5$.

- (A) $-3 < x < 7$
 (B) $-2 < x < 6$
 (C) $-5 < x < 3$
 (D) $0 < x < 4$
 (E) $1 < x < 5$

Q2. The absolute value inequality $|2x + 3| \leq 5$ is equivalent to:

- (A) $-4 \leq x \leq 1$
 (B) $-1 \leq x \leq 2$
 (C) $-2 \leq x \leq 3$
 (D) $-3 \leq x \leq 4$
 (E) $-5 \leq x \leq 6$

Q3. The temperature in a greenhouse varies between 65°F and 75°F . If t represents the temperature, which inequality best represents this situation?

- (A) $|t - 70| \leq 5$
 (B) $|t - 70| \geq 5$
 (C) $|t - 70| < 5$
 (D) $|t - 70| > 5$
 (E) $|t - 70| = 5$

Q4. The graph of the solution to $|x + 1| > 4$ is:

- (A) $x < -5$ or $x > 3$
 (B) $x < -3$ or $x > 5$
 (C) $-5 < x < 3$
 (D) $-3 < x < 5$
 (E) $x < -1$ or $x > 1$

Q5. The city of Miami has a rainfall of x inches per month. The amount of rainfall x satisfies $|x - 4| \leq 2$. This means that:

- (A) $2 \leq x \leq 6$
 (B) $2 < x < 6$
 (C) $0 \leq x \leq 8$
 (D) $4 < x < 8$
 (E) $0 < x < 8$

Q6. Solve the inequality $|3x - 2| < 11$.

- (A) $-3 < x < 4$
 (B) $-2 < x < 3$
 (C) $-1 < x < 2$
 (D) $0 < x < 5$
 (E) $1 < x < 6$

Q7. The absolute value inequality $|x - 5| \geq 2$ is equivalent to:

- (A) $x \leq 3$ or $x \geq 7$
 (B) $x < 3$ or $x > 7$
 (C) $3 \leq x \leq 7$
 (D) $3 < x < 7$
 (E) $x = 3$ or $x = 7$

Q8. If $|2x + 5| \leq 3$, then:

- (A) $-4 \leq x \leq 1$
 (B) $-4 \leq x \leq -1$
 (C) $-2 \leq x \leq 1$
 (D) $-1 \leq x \leq 2$
 (E) $-1 \leq x \leq -1$

Open Ended Questions (FRQ)

Q9. A certain species of birds can only survive in temperatures between 40°F and 80°F . If T represents the temperature in degrees Fahrenheit, write an absolute value inequality that represents the temperature range in which the birds can survive. Provide your answer in interval notation.**Q10.** The water level in a lake varies between 20 feet and 40 feet. If w represents the water level, write an absolute value inequality that represents this situation. Then, solve the inequality and provide the solution in interval notation.

Chef's Tips

- Tip 1: Emphasize the importance of understanding the concept of absolute value and its application in inequalities.
- Tip 2: Make sure students can graph the solution sets on a number line and express them in interval notation.
- Tip 3: Encourage students to use real-world examples, such as weather and environmental changes, to model absolute value inequalities.